

Rewriting Expressions with Distributive Structure

Answer Key

Algebra 1

Quick Answers

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|-----------------------------------|---|
| 1. $4x + 26$ m | 7. $25x - 100$ |
| 2. $15x + 45$ | 8. (a) $= x + 4$ m, — |
| 3. $x + 5$ m | 9. $8x + 5$ m |
| 4. (a) $= 8x + 40$, — | 10. (a) standard form $= 3x^2 + 18x + 24$ cm ² , (b) |
| 5. $x^2 + 8x + 12$ m ² | factored form $= 3(x + 2)(x + 4)$, — |
| 6. $3x(3x + 5)$ | |
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What is verified

7 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 4, 8, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-APR.A.1 — *problems 1, 2, 4, 5, 7, 9, 10*

HSA-SSE.B.3 — *problems 3, 6, 8, 10*

Difficulty 1–5 of 5 across 14 tagged checks.

- 1.** Dog run $(2x + 9)$ m by 4 m: perimeter in metres, no brackets.

Solution: Perimeter is two long sides plus two short sides, so $2(2x + 9) + 2(4)$. Distribute the 2 across both terms of the bracket: $2 \cdot 2x = 4x$ and $2 \cdot 9 = 18$, and $2 \cdot 4 = 8$. Then $18 + 8 = 26$.

$4x + 26$ m

- 2.** 15 notebooks at $(x + 3)$ dollars each: total cost in dollars, no brackets.

Solution: Fifteen copies of the same price is $15(x + 3)$. The 15 multiplies both parts of the price — the variable part and the fixed part: $15 \cdot x = 15x$ and $15 \cdot 3 = 45$. Read the answer back: the cost rises by 15 dollars for each extra dollar of notebook price, on top of a fixed 45 dollars.

$15x + 45$

- 3.** Hallway of area $(6x + 30)$ square metres and width 6 m: find the length in metres.

Solution: Area = width $\times$ length, so length = area $\div$ width. Taking the common factor 6 out of the area shows the division without long working: $6x + 30 = 6(x + 5)$, so the length is the second factor. $x + 5$ m

4. Plan A $8(x + 5)$ dollars, Plan B $8x + 40$ dollars: (a) confirm they agree; (b) what does each form show?

Solution (a): Distribute the 8: $8 \cdot x = 8x$ and $8 \cdot 5 = 40$, so Plan A charges $8x + 40$ dollars, the same as Plan B for every x . $8x + 40$

Solution (b) — instructor-judged. The factored form $8(x + 5)$ shows the price per book, 8 dollars, and the 5 spare books included in the order. The expanded form $8x + 40$ shows the flat 40-dollar charge for those spares as a separate term, which is what you want when comparing against a plan with a different fixed fee. Full credit connects each form to the quantity it makes visible; describing only one form earns half credit.

5. Patio $(x + 6)$ m by $(x + 2)$ m: area in square metres, standard form.

Solution:

$$\begin{aligned}(x + 6)(x + 2) &= x \cdot x + x \cdot 2 + 6 \cdot x + 6 \cdot 2 \\ &= x^2 + 2x + 6x + 12 = x^2 + 8x + 12.\end{aligned}$$

Each of the four products is one rectangle of the area model.

$x^2 + 8x + 12$ m²

6. Write $(9x^2 + 15x)$ square metres as a product of the greatest common monomial factor and one remaining factor.

Solution: The largest number dividing both 9 and 15 is 3, and the lowest power of x present in both terms is x , so the greatest common monomial factor is $3x$: $9x^2 + 15x = 3x \cdot 3x + 3x \cdot 5 = 3x(3x + 5)$. Re-distributing gives back $9x^2 + 15x$, which is the check.

$3x(3x + 5)$

7. Income $40x$ dollars, costs $5(3x + 20)$ dollars: profit in dollars, no brackets.

Solution: Profit = $40x - 5(3x + 20)$. Distribute the 5 first, and keep the minus sign with *both* products: $5(3x + 20) = 15x + 100$, so $40x - 15x - 100 = 25x - 100$. Subtracting only the $15x$ and forgetting the 100 is the usual slip here. $25x - 100$

8. Field $(3x + 12)$ m in 3 equal strips: (a) length of one strip; (b) why the factored form shows it.

Solution (a): One strip is a third of the length. Factoring first, $3x + 12 = 3(x + 4)$, so dividing by 3 leaves the second factor. $x + 4$ m

Solution (b) — instructor-judged. In the form $3(x + 4)$ the field is literally written as 3 copies of a length $x + 4$, so the number of strips is one of the factors on the page; finding one strip is reading off the other factor, not computing a division. Full credit refers to the factor 3 standing for the number of strips; a response that only recomputes the division earns half credit.

- 9.** Edging: $3(2x + 5) + 4(x - 2) - 2(x + 1)$ metres. Total length, no brackets.

Solution:

$$3(2x + 5) = 6x + 15,$$

$$4(x - 2) = 4x - 8,$$

$$2(x + 1) = 2x + 2 \quad (\text{this whole amount is removed}),$$

so the total is

$$6x + 15 + 4x - 8 - 2x - 2 = (6x + 4x - 2x) + (15 - 8 - 2) = 8x + 5.$$

$8x + 5 \text{ m}$

- 10.** Sheet $(3x + 6)$ cm by $(x + 4)$ cm: (a) standard form; (b) factored form; (c) which form shows divisibility by 3?

Solution (a):

$$\begin{aligned}(3x + 6)(x + 4) &= 3x \cdot x + 3x \cdot 4 + 6 \cdot x + 6 \cdot 4 \\ &= 3x^2 + 12x + 6x + 24 = 3x^2 + 18x + 24.\end{aligned}$$

$3x^2 + 18x + 24 \text{ cm}^2$

Solution (b): Every coefficient is divisible by 3, so take 3 out first: $3x^2 + 18x + 24 = 3(x^2 + 6x + 8)$. Then $x^2 + 6x + 8$ needs two numbers with product 8 and sum 6, namely 2 and 4.

$3(x + 2)(x + 4)$

Solution (c) — instructor-judged. The factored form. The leading factor 3 multiplies the whole product, so for any whole-number x the area is 3 times a whole number and is therefore divisible by 3. The standard form hides this: you would have to check each coefficient. Full credit names the visible factor 3 as the reason; answering only that both forms give the same number earns half credit.